

Attention Flags, MLPs Write: Mechanistic Lag in GPT-2 Adjective–Noun Binding

Sparse Autoencoders Reveal Residual Stream Feature Writing

ABSTRACT

Transformers are black boxes and we as researchers want to gain an understanding as to how models think for AI safety and uncovering novel understanding.

Research Question:

Does the residual stream representation of an object measurably acquire semantic features associated with a modifier, and if so, through what mechanism?

Ex. "A Greek king sat on his throne."

modifier

noun/object

KEY TERMS

Polysemancity:

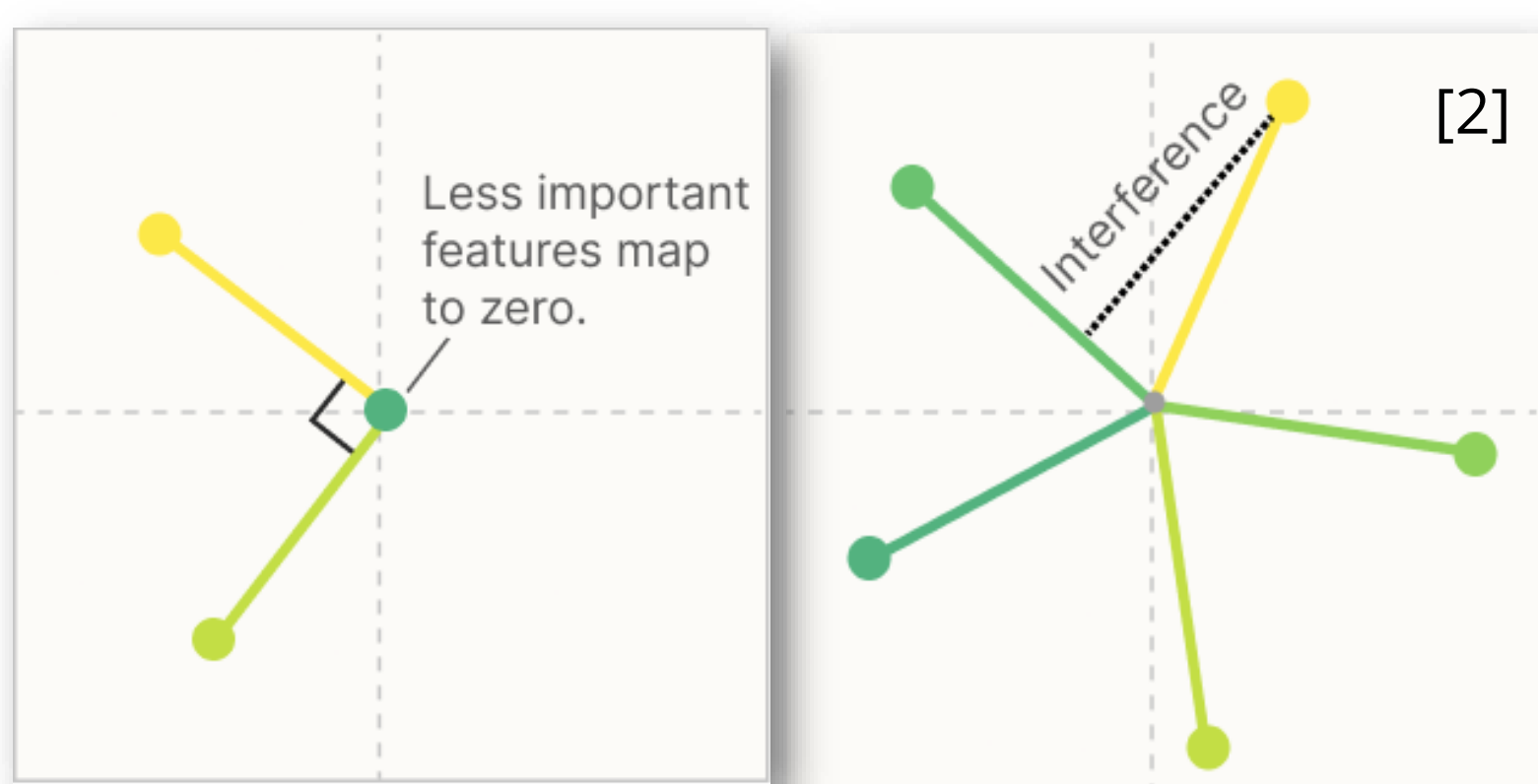
Individual neurons in neural networks respond to multiple unrelated concepts rather than cleanly encoding a single feature.

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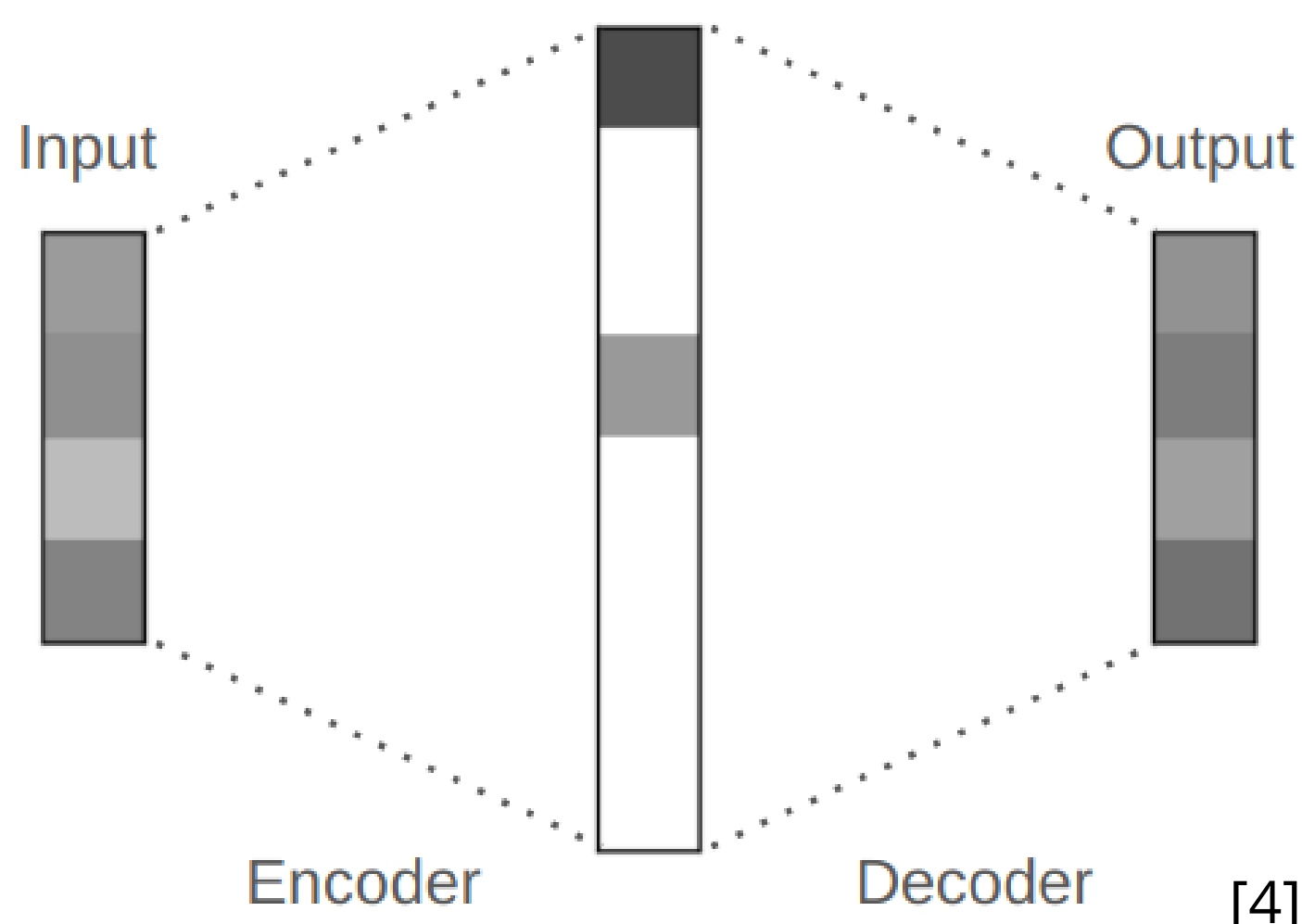


Superposition: Models represent more features than they have dimensions by storing many almost-orthogonal vectors in the same space, at the cost of interference.

Johnson-Lindenstrauss Lemma: A model with n dimensions can represent up to $O(\exp(n))$ near-orthogonal features.

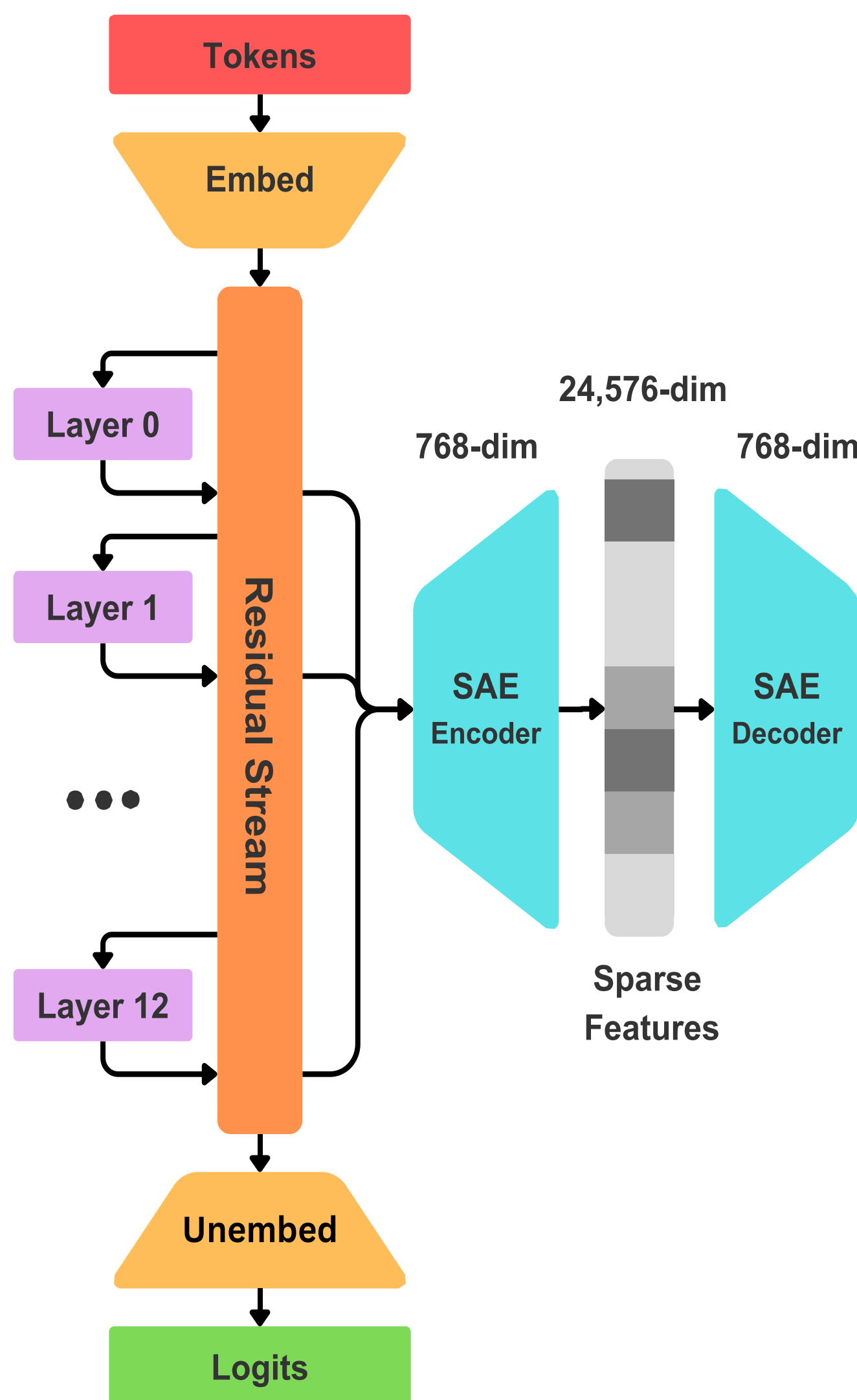


Sparse Autoencoders: A tool for decomposing a model's internal activations into a larger set of interpretable, sparse features that disentangle superposition.



METHODOLOGY

1. Tokenize & run forward pass
2. Extract noun residual stream per layer
3. Encode via same same-layer SAE
4. Compute Δ activations $\rightarrow$ identify modifier feature
5. Project through SAE decoder $\rightarrow$ direction d
6. Measure cosine similarity and adjective attention weights



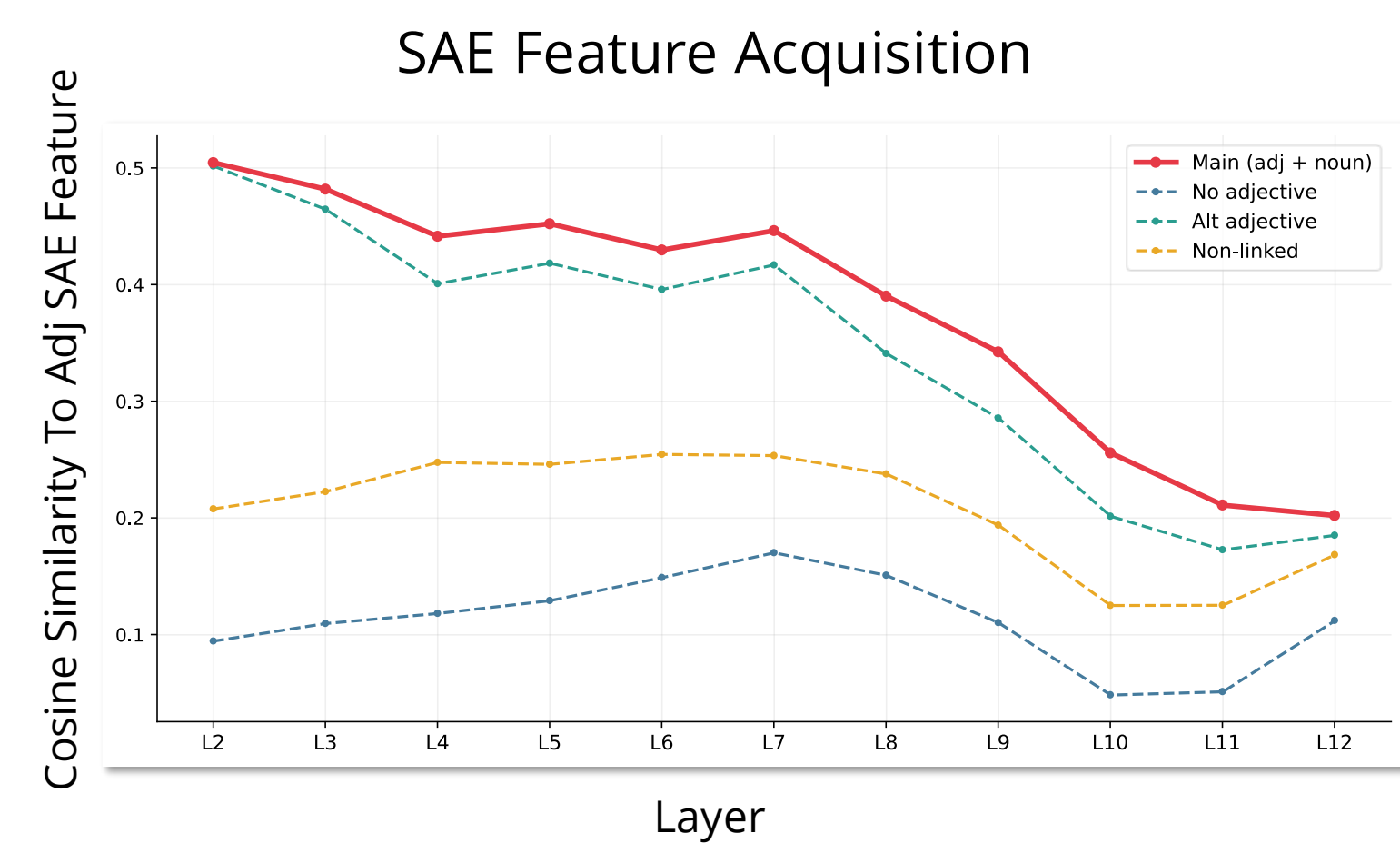
Additional Details:

- **Models:** GPT-2 Small (117M) · SAE: gpt2-small-res-jb (768-dim $\rightarrow$ 24,576-dim)
- **Data:** 200 adj-noun pairs, 4 conditions each
- **Control Groups:** main, no-adj, alt-adj, non-linked
- **Tools:** TransformerLens, SAE Lens

KEY TAKEAWAYS

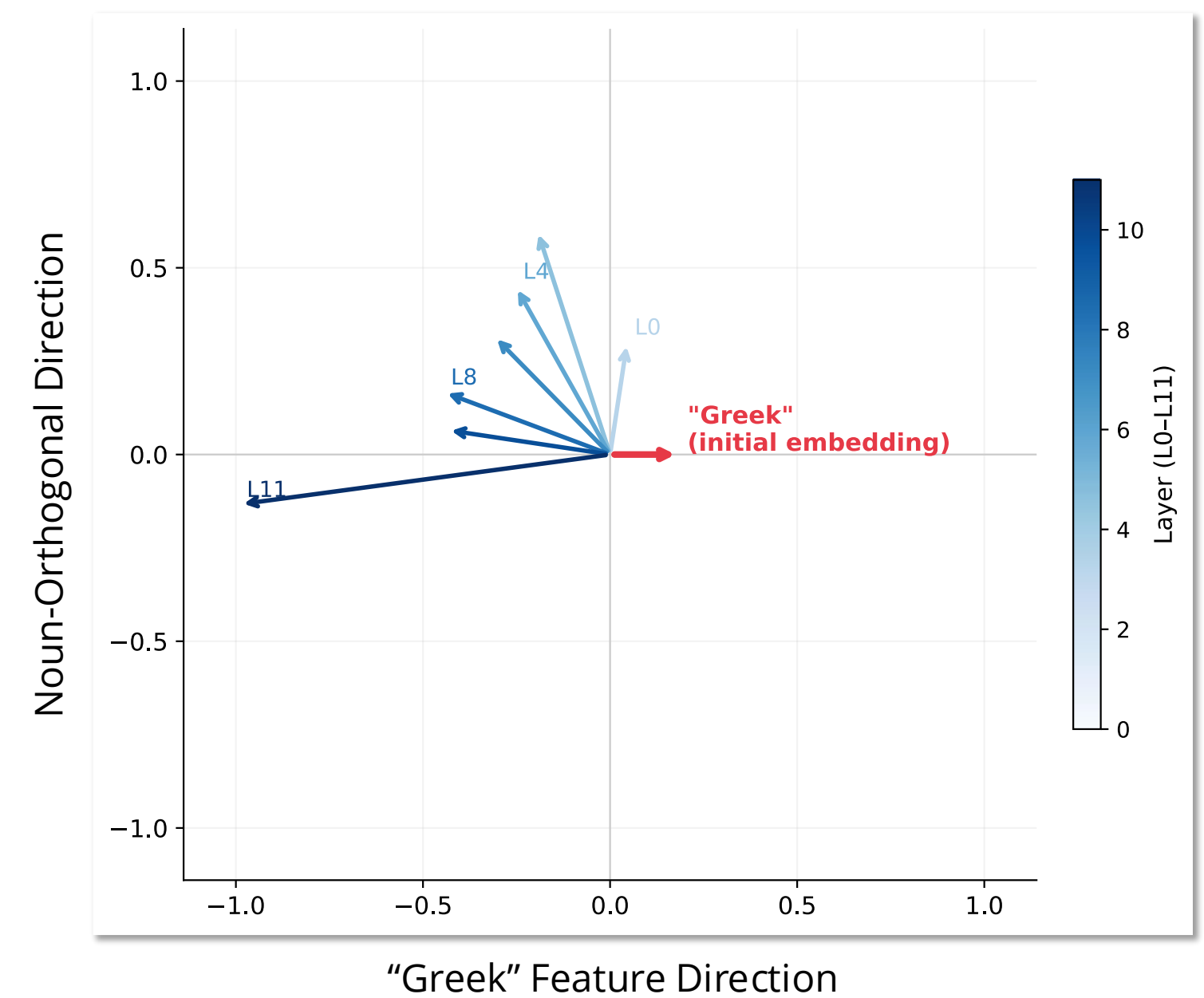
- Attention from noun $\rightarrow$ adjective usually peaks at Layer 2; SAE features integrate 1-5 layers later (mean 3.4), suggesting a **consistent two-stage mechanism: attention flags the relationship, MLPs write the features**
- After attention fires, **the noun's residual stream representation diverges from the raw adjective embedding**, confirming the model computes a transformed semantic representation rather than performing simple vector addition

RESULTS & ANALYSIS



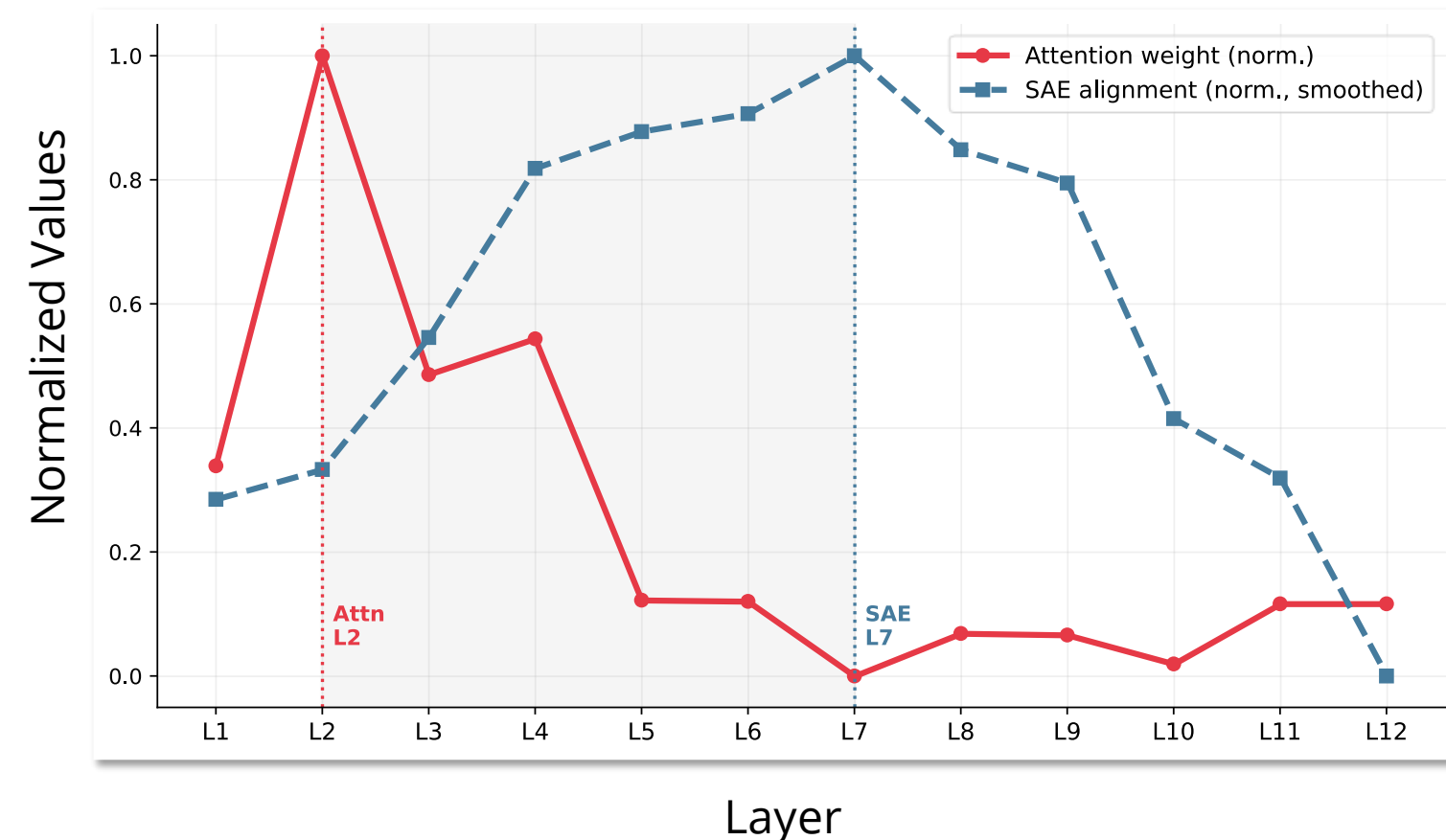
Main condition maintains noticeably higher cosine similarity with modifier feature than all controls across layers.

"King" Vector Trajectory vs. Layers



"King" vector rotates away from raw "Greek" embedding—**model computes, not copies.**

Mechanistic Lag Between Attention & SAE Feature Activation



Attention peaks at L2; SAE features peak L5–L10 (**mean 3.4**) across all 200 pairs.

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